

Deflectometry Mirror-Array Mathematical Reference

*Array sizing, fringe coding, SMOTS tilt retrieval,
error budget and screen-pose calibration*

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Companion reference to the Deflectometry Mirror-Array Toolkit and the `smots` Python package.

Every equation here is implemented and exercised by the toolkit; section headings name the file that implements it.

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1. Notation and coordinate frames

Read this page first. Two of the three sign conventions below are places where a printed equation and a standard software convention disagree, and each one silently inverts an angle.

1.1 Coordinate frames

The mirror array lies in the plane $z = 0$ with nominal surface normals along $+z$. The screen is a plane in front of it, nominally parallel, at perpendicular distance z_d . The camera looks back along $-z$. Node labels run in reading order, so the centre of a 3×3 array is M5 — matching the mechanical assembly drawings.

A tilt component is named after the screen axis it moves: θ_x rotates the surface normal toward $+x$ and therefore shifts the observed pattern along the screen's x axis. Keeping that naming consistent end to end is what lets you put θ_x in and get θ_x back out.

1.2 The three sign conventions

Where	What is written	What this work uses
Fourier shift theorem	Eq. (5.2) is printed as $e^{-i2\pi f\Delta F(f)}$	The standard forward transform gives $e^{+i2\pi f\Delta F(f)}$ for that same shift. Positive phase here means the pattern moved toward $+x$.
Tilt from screen shift	Eq. (5.12) as printed subtracts the final arctangent from the initial one	That order returns a <i>negative</i> angle for a normal rotating toward $+x$. The order is reversed here so the sign reads naturally.
Image row direction	—	Camera frames store row 0 at the top, so the row index runs <i>against</i> world $+y$. The y shift is negated. Omit this and x reads correctly while y inverts.

Why this matters. None of the three changes any magnitude, so no residual, no RMS and no consistency check will catch them. A mirror tilted one way is simply reported as tilted the other. Confirm the signs against a commanded actuator move before trusting any measurement.

2. The bench

The as-built 9-node hybrid assembly: a 3×3 array of 1" square mirrors, the centre node motorised, the other eight hand-set references.

Quantity	Value	Note
Mirror	25.4 mm square	PFSQ10-03-G01, $\lambda/10$ surface flatness
Array	3×3 , 3 mm gap	82.2×82.2 mm overall
Mount	KMSR	$\varnothing 1"$ kinematic, one actuator per tilt axis
Actuator	$2 \times$ MPIA10	Centre node M5 only; piezo inertia, tip and tilt
Controller	KIM101	4 channels; a tip/tilt node uses 2, so 2 nodes maximum
Reference nodes	TR1.5 / PH1.5	Eight peripheral nodes, height set by hand
Screen	344 mm, 3840 px	Nominal pitch $89.6 \mu\text{m}$

Unverified inputs. Five numbers scale the result and are *not* datasheet values in the toolkit: the screen pixel pitch, the mirror-to-screen distance z_d , the mount lever arm L , the actuator travel and the actuator step size. The first two scale every reported angle directly — § 5.6 and § 5.7 show how. Measure them; do not inherit them from this document.

3. Array configuration

Sizing the array against the screen and the camera. Implemented in `index.html`.

3.1 Screen pixel pitch

$$p = W / N_x \quad (3.1)$$

W screen width · **N_x** horizontal pixel count. Nominally 344 mm / 3840 = 89.6 μm. This number multiplies every angle the system reports (§ 5.6), which is why it is worth measuring under a microscope rather than dividing datasheet figures.

3.2 Fringe pitch

$$P_f = n \cdot p \quad (3.2)$$

n pixels per fringe. The physical period of the pattern on the screen, in mm.

3.3 Array extent

$$L = n_c a + (n_c - 1) g \quad (3.3)$$

n_c columns · **a** mirror side · **g** inter-mirror gap. The same form gives the height from the row count.

3.4 Fringes across the array

$$N_f = L / P_f \quad (3.4)$$

Target **N_f** ≥ 60 for reliable phase extraction; below about 40 the measurement degrades. This is the headline sizing constraint.

3.5 Screen angular field

$$\theta = 2 \arctan(W_s / 2d) \quad (3.5)$$

The angular extent the screen subtends from the array at distance *d*. The array must fall inside it, or the outer mirrors reflect past the panel edge.

3.6 Camera angular field

$$\text{AOV} = 2 \arctan(s / 2f) \quad \text{FOV}(d) = 2d \tan(\text{AOV}/2) \quad (3.6)$$

s sensor dimension · **f** focal length. The covered field at the mirror plane must exceed the array extent from (3.3), else the outer nodes fall outside the frame.

3.7 Surface flatness

$$\Delta z \leq \lambda/4 \rightarrow \leq 137 \text{ nm} \quad (\text{actual} \leq 55 \text{ nm at } \lambda/10) \quad (3.7)$$

A property of the mirrors, not of the measurement. It sets the floor the measurement is trying to certify; the measurement's own error must sit well below it (§ 7.8).

4. Fringe generation and phase-shift coding

What is displayed on the screen, and what a conventional N -step measurement extracts from it. Implemented in `fringes.html`.

4.1 Projected intensity

$$I_k(x) = A + B \cos(2\pi x/n - 2\pi k/N) \quad (4.1)$$

A bias (mean grey level) · **B** modulation amplitude · **k** = 0 ... $N-1$ the step index. Requires $A \pm B$ to stay inside 0–255: a clipped sinusoid is no longer sinusoidal, and the clipping biases the retrieved phase rather than merely adding noise.

4.2 N -step phase retrieval

$$\varphi = \arctan2(\sum_k I_k \sin(2\pi k/N), \sum_k I_k \cos(2\pi k/N)) \quad (4.2)$$

Wrapped to $(-\pi, \pi]$. This is the classical approach and it needs N captured frames per measurement. SMOTS (§ 5) does not use it — that is the central difference between the two methods and the reason SMOTS is fast.

4.3 Phase uncertainty

$$\sigma_\varphi = \sqrt{(2/N)} \cdot \sigma_I / B / \sqrt{M} \quad (4.3)$$

σ_I per-pixel intensity noise in grey levels · **M** frames averaged. More steps, more modulation and more averaging all help, and all cost time. This feeds the error budget as its only genuinely random term.

4.4 Screen-position error from phase noise

$$\delta x_s = \sigma_\varphi \cdot P_f / 2\pi \quad (4.4)$$

A coarser fringe turns the same phase noise into a larger position error, which is the resolution half of the resolution-versus-range trade that § 5.9 formalises.

4.5 Gamma pre-compensation

$$I_{\text{out}} = 255 \cdot (I_{\text{lin}} / 255)^{1/\gamma} \quad (4.5)$$

Panels apply a transfer curve of roughly $\gamma \approx 2.2$. Pre-compensating makes the *emitted irradiance* sinusoidal rather than the commanded grey level. Left uncorrected, γ injects a second harmonic that lands directly in the retrieved phase.

5. SMOTS: sheared Fourier tilt retrieval

The method this research is built on. One pattern is displayed and never moved; one reference frame is captured; every later frame is compared against it. Implemented in `smots/analysis.py` and `smots/geometry.py`, after Choi et al. [1].

The idea in one sentence. Subtract the reference image from the measured image, and the amplitude of what is left tells you how far the pattern moved — because subtracting two copies of the same sinusoid offset by Δ leaves a sinusoid whose amplitude grows with Δ .

5.1 The shear theorem

$$\text{FT}[f(x)] = F(f) \quad (5.1)$$

$$\text{FT}[f(x+\Delta)] = e^{i\psi} F(f), \quad \psi = 2\pi f\Delta \quad (5.2)$$

$$S(f) = \text{FT}[f(x+\Delta) - f(x)] = (e^{i\psi} - 1) F(f) \quad (5.3)$$

A shift is a phase ramp in the Fourier domain, so the *sheared* pattern — the difference of the two frames — has the same frequency as the original but an amplitude set by the shift. Everything else follows from (5.3).

5.2 Phase from the magnitude ratio (as published)

$$|S/F|^2 = 2 - 2 \cos \psi \rightarrow \psi = \arccos[1 - \frac{1}{2} |S/F|^2] \quad (5.4)$$

Uses only magnitudes, so it returns an *unsigned* angle in $[0, \pi]$. The direction has to be recovered separately, from the sign of the imaginary part.

5.3 Phase from the complex ratio (recommended)

$$S/F = e^{i\psi} - 1 \rightarrow \text{Re} = \cos \psi - 1, \quad \text{Im} = \sin \psi \quad (5.5)$$

$$\psi = \arctan2(\text{Im}(S/F), 1 + \text{Re}(S/F)) \quad (5.6)$$

Why prefer (5.6). It is signed, spans the full $(-\pi, \pi]$ without a separate sign step, and stays well conditioned where (5.4) does not. The derivative of $\arccos$ is unbounded at $\psi \rightarrow 0$ and $\psi \rightarrow \pi$ — precisely where small shifts and near-wrapping shifts live. Both are implemented; they agree away from those corners, and the test suite checks that they do.

5.4 Locating the carrier

The carrier is found in the camera's own spectrum rather than assumed. Its bin index is *aperture / period*, both measured in the same units, so the camera magnification never enters — which is what makes the method robust to an uncalibrated imaging scale.

5.5 Phase to length on the screen

$$\Delta = (\psi / 2\pi) \cdot P_f \quad (5.7)$$

A phase *fraction of a period* is invariant under magnification; multiplying by the physical period on the screen converts it to millimetres. This is the step where the screen pixel pitch (3.1) enters, and the only place it does.

5.6 Screen point seen through the mirror

$$x_f = x_i + \Delta \quad (5.8)$$

x_i the screen point seen in the reference frame · x_f the point seen now.

5.7 Tilt angle

$$\theta = \frac{1}{2} [\arctan((x_f - x_0) / z_d) - \arctan((x_i - x_0) / z_d)] \quad (5.9)$$

x_0 the mirror's own position along the axis · z_d mirror-to-screen distance. The factor of $\frac{1}{2}$ is the physics: rotating a mirror by θ swings the reflected ray by 2θ , so the observed motion is twice the angle you are measuring.

5.8 Paraxial form

$$\theta \approx \Delta / (2 z_d) \quad (5.10)$$

Valid near the axis and for small angles; agrees with (5.9) to better than 0.1 % for a mirror within a few percent of z_d off axis. Useful as an independent check that the full form is wired up correctly.

5.9 Resolution and unambiguous range

$$\delta\theta = (\sigma_\phi / 2\pi) P_f / (2 z_d) \quad \theta_{\max} = \pm P_f / (4 z_d) \quad (5.11)$$

The trade in one line: a finer fringe buys resolution and costs range, in exact proportion. The shift wraps at $\pm P_f/2$, hence the factor 4.

5.10 Breaking the 2π ambiguity

Two carriers of different period are displayed at once. Each reports its shift only modulo its own period, and the true shift is the one both agree on. The pair realigns at the *least common multiple* of the two periods, so consistent solutions repeat forever at that spacing — taking the globally best-agreeing candidate returns an arbitrary alias. The correct rule is to keep every candidate that agrees within tolerance and select the one nearest a prior (zero, or the previous measurement).

$$\Lambda_{\text{beat}} = P_1 P_2 / | P_1 - P_2 | \quad (5.12)$$

Beat period is not the repeat. Writing $P_2/P_1 = p/q$ in lowest terms, solutions actually repeat at $pP_1 = (p-q) \Lambda_{\text{beat}}$. So a 4:3 pair repeats exactly at the beat, while a 5:3 pair repeats at twice it. Quoting the beat under-claims the range, which is the safe direction.

Choosing the periods is a real trade: closer periods lengthen the range, but a carrier's FFT bin index is *aperture/period*, so nearly equal periods land in nearly the same bin and cannot be separated at all.

5.11 Digital apertures

A binary mask is applied per mirror segment, and the analysis runs independently inside each. This is what makes the measurement *simultaneous*: every segment is its own small interferogram inside one camera frame, and all of them are solved from the same pair of images. The averaged result within an aperture is analogous to a Shack–Hartmann lenslet averaging the wavefront over its own footprint.

$$\text{aperture width} / P_f \geq 3 \text{ periods}$$

(5.13)

Below roughly three periods across an aperture the Fourier peak is too broad to locate reliably, and the segment should be treated as unmeasured rather than trusted.

5.12 Common-mode rejection

$$\Delta\theta_i = \theta_i - \text{median}_j(\theta_j)$$

(5.14)

Watching every segment at once means a global disturbance — the bench drifting, someone leaning on the table — appears as common motion across all of them and can be subtracted, leaving the genuine relative change on the node that actually moved.

Relative, not absolute. Every angle from (5.9) is measured *against the reference capture*. Absolute orientation requires a separate absolute reference, which is why the original work cross-checked against an autocollimator. Screen-pose calibration (§ 8) likewise needs absolute screen coordinates, so unwrapped phase, not the relative shift.

6. Mirror tilt and piezo actuation

Where each mirror has to point, and whether the hardware can get it there. Implemented in [piezo.html](#).

6.1 Incident and reflected directions

$$\hat{u} = (\mathbf{M} - \mathbf{C}) / \|\mathbf{M} - \mathbf{C}\| \quad \hat{v} = (\mathbf{S} - \mathbf{M}) / \|\mathbf{S} - \mathbf{M}\| \quad (6.1)$$

$\mathbf{C}$ camera · $\mathbf{M}$ mirror centre · $\mathbf{S}$ screen centre. $\hat{u}$ runs from camera to mirror, $\hat{v}$ from mirror to screen.

6.2 Law of reflection, solved for the normal

$$\hat{v} = \hat{u} - 2(\hat{u} \cdot \hat{n})\hat{n} \Rightarrow \hat{v} - \hat{u} = -2(\hat{u} \cdot \hat{n})\hat{n} \Rightarrow \hat{n} \propto \hat{v} - \hat{u} \quad (6.2)$$

The usual statement of reflection is a forward model; rearranged, it inverts directly. The surface normal *bisects* the incoming and outgoing rays, so a mirror's required orientation follows in closed form — no iteration. The sign of $\hat{n}$ is fixed by requiring it to face the camera.

6.3 Tilt components from the normal

$$\theta_x = \arctan(n_x / n_z) \quad \theta_y = \arctan(n_y / n_z) \quad \theta_{\text{tot}} = \arccos(n_z) \quad (6.3)$$

6.4 Splitting the baseplate from the piezos

$$\theta_{\text{base}} = \text{mean}_i(\theta_i) \quad \theta_{\text{piezo},i} = \theta_i - \theta_{\text{base}} \quad (6.4)$$

This split is the design. Every mirror needs a large, nearly identical fold angle to route the camera onto the screen, plus a small differential that varies across the array. The common part is set once, mechanically, by how the baseplate is bolted. Only the differential has to come from the actuators. Charging the piezos for the whole fold overstates their job by several times and makes an otherwise feasible design look impossible.

6.5 Actuator displacement

$$\Delta = L \tan \theta_{\text{piezo}} \quad (6.5)$$

L lever arm from mount pivot to actuator. One actuator per axis.

6.6 Angular resolution

$$\delta\theta = \delta\Delta / L \quad \delta s = 2 \delta\theta d \quad (6.6)$$

One actuator step of $\delta\Delta$ tilts the mirror by $\delta\theta$ and walks the reflected spot δs on the screen — again the factor 2 from § 5.7. Compare δs against the fringe pitch to see whether a single step is even resolvable.

6.7 Range budget

$$\theta_{\max} = \min[\theta_{\text{mount}} , \arctan(T / 2L)] \quad (6.7)$$

T total actuator travel. Whichever limit binds first is the real one. A motorised node is capped by both; a hand-set reference node only by the mount's own adjuster range, since nothing is driving it.

6.8 Controller channel budget

$$N_{\text{motorised}} \leq \lfloor \text{channels} / 2 \rfloor \quad (6.8)$$

A tip/tilt node consumes two channels. A KIM101 offers four, so **one controller drives at most two nodes** — a hard constraint on how much of the array can ever be actuated.

7. Measurement error budget

Propagating every screen-side uncertainty into a slope error, then into a height error. Implemented in `budget.html`.

7.1 Slope sensitivity

$$\delta s = \delta x_s / (2d) \quad (7.1)$$

A slope change δs rotates the reflected ray by $2\delta s$, walking the screen point by $2\delta s d$. Inverting gives the sensitivity: every screen-position error divides by $2d$ to become a slope error. Longer standoff is free accuracy here.

7.2 Contributors

Term	Slope error	Character
Phase noise	$\delta x_s / 2d$, from (4.4)	random — uncorrelated pixel to pixel
Screen pose	$\delta x_{\text{cal}} / 2d$	tilt-like — uniform across the aperture
Screen planarity	$\delta x_{\text{plan}} / 2d$	figure — correlated across the aperture
Distance error	$s \cdot \delta d / d$	scale — proportional to the true slope

7.3 Quadrature sum

$$\delta s_{\text{RSS}} = \sqrt{(\sum_i \delta s_i^2)} \quad (7.2)$$

Valid while the terms are independent. A common-mode calibration error shared between them would add linearly instead, and this would understate it.

7.4 Integration into height — one rule per character

$$\text{random: } \delta z = \delta s \sqrt{L \cdot \Delta x} \quad (7.3)$$

$$\text{figure: } \delta z = \delta s \cdot L \quad (7.4)$$

$$\text{tilt-like: } \delta z = 0 \text{ (removed by the piston-and-tilt fit)} \quad (7.5)$$

$$\text{scale: } \delta z = \text{sag} \cdot \delta d / d \quad (7.6)$$

Do not integrate every term the same way. Slope errors do not share a single propagation rule, and treating them as though they do is a common and expensive mistake. Random noise averages down as a random walk over the aperture. Correlated figure error does not average at all and integrates coherently — a penalty of $L/\sqrt{L\Delta x}$, which for this bench is a factor of about fifty. A uniform screen-pose offset gives every point the *same* slope error, which is a pure tilt and is therefore removed *exactly* by the reconstruction's low-order fit — it contributes nothing, not merely a little.

7.5 Lateral registration

$$\delta z_{\text{reg}} = s \cdot \delta x_{\text{reg}} \quad (7.7)$$

Attributing a measured slope to the wrong point on the surface costs height in proportion to the local slope itself.

7.6 Against the surface specification

$$\delta z_{\text{tot}} = \sqrt{(\sum_i \delta z_i^2)} \quad \text{vs} \quad \lambda/10 \quad (7.8)$$

Slope is the honest deliverable. Deflectometry measures slope *directly*; height is an integrated, assumption-dependent quantity. Reaching $\lambda/10$ in absolute height demands the screen geometry be known to a few micrometres, which is a far harder requirement than reaching good slope accuracy. Quote slope as the primary result and treat height as what it costs you if the geometry is only known as well as it is.

8. Screen-pose calibration through the mirrors

Where z_d should come from. Implemented in `smots/calibration.py`; formulation developed from the geometry for this work, structured after the conventional linear-then-nonlinear approach [2].

The camera never sees the screen — it looks at the mirrors. So the screen's pose has to be recovered *through* the reflection. This matters because § 7 finds screen geometry dominating the whole error budget, and (5.9) needs z_d .

8.1 Plane reflection

$$\mathbf{H}(\hat{\mathbf{n}}, e) : \quad \mathbf{R} = \mathbf{I} - 2 \hat{\mathbf{n}} \hat{\mathbf{n}}^T, \quad \mathbf{t} = 2 e \hat{\mathbf{n}} \quad (8.1)$$

The Householder reflection through the plane $\{ \mathbf{x} : \hat{\mathbf{n}} \cdot \mathbf{x} = e \}$. It is an involution — reflecting twice returns the identity — and its determinant is -1 , so it flips handedness. What the camera sees through a flat mirror is a *virtual* screen, and the real screen is that virtual one reflected back through the mirror plane.

8.2 The ray-correspondence constraint

Each observation is a triple: a point $\mathbf{P}$ on a mirror of known normal, the direction $\hat{\mathbf{r}}$ the camera ray travels after reflecting there, and the screen coordinate $\mathbf{q} = (u, v)$ that the decoded absolute pattern says is seen at that point. The screen point must lie somewhere on that reflected ray:

$$\mathbf{R}\mathbf{q} + \mathbf{t} = \mathbf{P} + \lambda \hat{\mathbf{r}} \quad (8.2)$$

$$(\mathbf{I} - \hat{\mathbf{r}} \hat{\mathbf{r}}^T) (\mathbf{R}\mathbf{q} + \mathbf{t} - \mathbf{P}) = 0 \quad (8.3)$$

Projecting perpendicular to the ray eliminates the unknown length λ , leaving two independent constraints per observation on the six-DOF pose.

8.3 Linear solution

Since $\mathbf{q}$ always has zero third component, the data constrains only the first two columns of $\mathbf{R}$. Solve the nine unknowns $[\mathbf{r}_1, \mathbf{r}_2, \mathbf{t}]$ by least squares, orthonormalise $[\mathbf{r}_1 \ \mathbf{r}_2]$ by SVD, and take $\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2$.

8.4 Nonlinear refinement

$$\min_{\mathbf{R}, \mathbf{t}} \sum_i \| (\mathbf{I} - \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^T) (\mathbf{R}\mathbf{q}_i + \mathbf{t} - \mathbf{P}_i) \|^2 \quad (8.4)$$

Gauss–Newton over all six degrees of freedom, with the rotation incremented through a Rodrigues parameterisation. This removes the bias the orthonormalisation step introduces.

8.5 Conditioning — and the degenerate case

$$\kappa = \lambda_{\min} [\sum_i (\mathbf{I} - \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^T)] / N \quad (8.5)$$

A perfectly flat array cannot be calibrated at all. Each observation constrains the screen only *across* its ray, never along it. If every mirror normal is identical then every reflected ray is parallel, the sum in (8.5) is singular, and z_d is not poorly determined — it is *not determined*, for any quantity of data.

The remedy is counter-intuitive: **do not set the reference mirrors parallel.** A few degrees of deliberate spread makes the depth observable. Measured on the synthetic bench with 50 μm decode noise, the z_d error falls from unobservable at 0° , to 0.345 mm at 0.5° , 0.078 mm at 2° , and 0.018 mm at 8° . Stepping only the motorised node helps far less, because it is one node in nine.

8.6 The screen pitch measures itself

$$k = \frac{1}{2} (\|\mathbf{r}_1\| + \|\mathbf{r}_2\|) \quad (\text{should be } 1) \quad (8.6)$$

A free diagnostic. If $\mathbf{q}$ is expressed in millimetres, the recovered screen axes must come out unit length. They do not if the assumed pixel pitch is wrong, and the deviation is the scale error — on exactly the number (3.1) that has to be measured under a microscope. The calibration therefore checks its own most important input.

8.7 Extracting what Eq. (5.9) needs

$$z_d = | (\mathbf{t} - \mathbf{o}) \cdot \hat{\mathbf{n}}_{\text{screen}} | \quad \text{tilt} = \arccos | \hat{\mathbf{n}}_{\text{array}} \cdot \hat{\mathbf{n}}_{\text{screen}} | \quad (8.7)$$

z_d is the *perpendicular* distance from the array origin to the screen plane — not the distance to the screen's origin, which for a tilted screen is a larger number. The tilt says how far the screen is from parallel; (5.9) assumes parallel, so any non-zero tilt is an error term otherwise absorbed silently into every reported angle.

Two prerequisites. Calibration needs *absolute* screen coordinates, so unwrapped phase — the relative SMOTS measurement never needs to know which fringe it is on, but this does. It also needs mirror normals in advance, which is circular since normals are what SMOTS measures. Break the loop with the mechanical normals of the reference nodes, which are set once and stable, or with an autocollimator; then calibrate; then measure.

9. Sign conventions: three traps, collected

Each of these was found by a numerical test disagreeing with a ray-traced forward model, not by reading an equation. Each inverts an angle while leaving every magnitude, residual and consistency check untouched.

Trap	Symptom	Resolution
Transform sign	Every recovered shift has the wrong sign. Magnitudes are perfect, so nothing flags it.	Eq. (5.2) as printed uses the opposite exponent sign to the standard forward transform. Eq. (5.4) is immune since it takes a magnitude; (5.6) is not. Fix the convention once, then test it.
Arctangent order	All tilts report with reversed polarity: a mirror tilted one way reads as tilted the other.	Eq. (5.9) as printed subtracts the final term from the initial one, returning a negative angle for a normal rotating toward +x. Reverse the order, or negate.
Image row direction	x is perfect and y is inverted — which looks exactly like a swapped actuator cable.	Row 0 is the top of the frame, so the row index runs against world +y. Negate the y shift once, at the point where image indices become world coordinates.

How to check on the bench. Command a known actuator move on the motorised node and confirm the reported angle changes in the expected direction and by roughly the expected amount. One deliberate move validates all three at once, and nothing else will.

10. Symbol glossary

Symbol	Units	Meaning
p	mm	Screen pixel pitch, Eq. (3.1). Scales every reported angle — measure it.
n	px	Pixels per fringe
P_f	mm	Fringe pitch on the screen, Eq. (3.2)
L	mm	Array extent (§3) or mount lever arm (§6) — context distinguishes them
N_f	—	Fringes across the array; target ≥ 60
N	—	Number of phase steps
A, B	DN	Bias and modulation of the projected sinusoid
σ_I	DN	Per-pixel intensity noise
σ_φ	rad	Phase uncertainty, Eq. (4.3)
ψ	rad	Sheared-pattern phase, Eq. (5.6)
Δ	mm	Pattern shift on the screen, Eq. (5.7)
F, S	—	Fourier transforms of the reference and sheared patterns
z_d	mm	Mirror-to-screen perpendicular distance. Angle goes as $1/z_d$.
θ	rad	Mirror tilt; reflected ray moves by 2θ
$\hat{n}$	—	Mirror surface normal, Eq. (6.2)
$\hat{u}, \hat{v}$	—	Incident and reflected unit directions
T	mm	Total actuator travel
s	rad	Surface slope
δs	rad	Slope error — the primary figure of merit
δz	mm	Height error after integrating slope
Δx	mm	Lateral sampling at the mirror
λ	nm	Wavelength; surface spec quoted as $\lambda/10$
R, t	—, mm	Screen pose: rotation and translation, §8
q	mm	Absolute screen coordinate seen through a mirror
κ	—	Ray conditioning, Eq. (8.5); zero means depth is unobservable

11. References

1. H. Choi, I. Trumper, M. Dubin, W. Zhao and D. W. Kim, “Simultaneous angular alignment of segmented mirrors using sinusoidal pattern analysis,” *Proc. SPIE* **10377**, 103770G (2017). — the source of §5.
2. D. Uhlig, *Light Field Imaging for Deflectometry*. — ch. 4 on phase coding and unwrapping, ch. 5 on system calibration. §8 here is structured after its linear-then-nonlinear pattern but was developed independently from the geometry; the text itself was not consulted in deriving it.
3. H. Choi, I. Trumper, M. Dubin, W. Zhao and D. W. Kim, “Simultaneous multi-segmented mirror orientation test system using a digital aperture based on sheared Fourier analysis,” *Opt. Express* **25**, 18152–18164 (2017).
4. P. Su, R. E. Parks, L. Wang, R. P. Angel and J. H. Burge, “Software configurable optical test system: a computerized reverse Hartmann test,” *Appl. Opt.* **49**, 4404–4412 (2010).
5. I. Trumper, H. Choi and D. W. Kim, “Instantaneous phase shifting deflectometry,” *Opt. Express* **24**, 27993–28007 (2016).
6. W. H. Southwell, “Wave-front estimation from wave-front slope measurements,” *J. Opt. Soc. Am.* **70**(8), 998–1006 (1980). — background for §7.4.

Companion material. Every equation above is implemented and exercised in the toolkit. The interactive pages ([index.html](#) , [fringes.html](#) , [piezo.html](#) , [budget.html](#) , [simulator.html](#)) evaluate them live; the `smots` Python package implements §5 and §8 with a ray-traced forward model and 63 tests that check the retrieval against known ground truth.